Fiscal Year:	FY 2020	**Task Last Updated:**	FY 02/17/2020
PI Name:	Hammond, Timothy M.D.		
Project Title:	Fuel to Mars		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-Artemis1 (EM1). App A: Orion (Artemis-1) (formerly Exploration Mission-1)
Start Date:	03/18/2019	**End Date:**	03/17/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	February 2020 report: None		
COI Name (Institution):	Allen, Patricia M.S. (Veterans Health Administration) Birdsall, Holly M.D., Ph.D. (Institute For Medical Research, Inc.) Giaever, Guri Ph.D. (University of British Columbia, Canada) Nislow, Corey Ph.D. (University of British Columbia, Canada)		
Grant/Contract No.:	80NSSC19K0706		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The primary research product this project will produce is identification of the insertion strains of Chlamydomonas reinhardtii, a single-cell green alga, which optimizes survival during exposure to the combined impacts of space radiation and microgravity in the cosmic space environment beyond the Van Allen Belts. The Chlamydomonas reinhardtii strains in a mapped insertional mutant library are a transformative tool allowing near genome wide analysis of the genes mediating survival advantage during stimuli such as cosmic radiation and microgravity. The secondary set of research products this project will produce are identification of the pathways mediating the gene changes, visualization of the pathway interactions, and identification of any environments giving similar gene pathway signatures in other model systems. Our goal is to identify strains of Chlamydomonas reinhardtii with gene insertions, which convey the best survival advantage during the exposure to the combined impacts of space radiation and microgravity for future use as parent strains in hydrogen production initiatives. Hydrogen is the lead contender for fuel production for return space flight missions from orbital bodies beyond low Earth orbit. The metabolism of select microorganisms makes hydrogen fuel. The best-known and most efficient hydrogen producer is Chlamydomonas reinhardtii, which is poised for exploitation as an industrial biotechnology platform for biofuel production. The type of investigation we propose is experiments that will be conducted aboard the Exploration Mission-1 (EM-1) [Note now called Artemis] lunar orbit flight, with asynchronous controls in Kennedy Space Center ISSES (International Space Station Environmental Simulators) Chambers. The proposed hardware is the NASA Biological Research in Canisters (BRIC)-100VC. Hypothesis: The hypothesis to be tested is that the mapped insertional mutant library of Chlamydomonas reinhardtii will allow identification of the genes that mediate optimal radiation resistance, and imperviousness to microgravity, in the cosmic space environment beyond the Van Allen Belts. The specific aims of the proposal are: Specific Aim 1: To experimentally determine which of the thousands of mapped insertion mutant strains of Chlamydomonas reinhardtii have the best long-term survival advantage during combined space radiation and microgravity impacts in the cosmic space environment beyond the Van Allen Belts. Specific Aim 2: To analyze the dataset of insertion strains with the best long-term survival advantage during combined space radiation and microgravity impacts to: • Perform pathway analysis using gene set enrichment analysis (GSEA); • Visualize the pathway interactions using Cytoscape/ClueGo; • Make comparison to homologous gene sets in Chlamydomonas, Saccharomyces cerevisiae, and other model organism databases for any environments giving similar gene pathway signatures. Although focused on gene pathways for survival for future applications in hydrogen fuel production, the data generated will provide genome level insights into the cancer risks of prolonged exposure to cosmic radiation and microgravity. The spectrum of cancer risk genes is known, and this study will parse the importance of risk genes, and the patterns of radiation risk change with exposure to the cosmic space environment beyond the Van Allen Belts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research impacts and Earth benefits include: o The first opportunity of apply modern “omics” techniques to samples flown beyond the van Allen belts, and recovered for ground-based analysis. o The first biological validation of the utility of the Chlamydomonas deletion series for academic and commercial applications. o Delineation of the production of the potential fuels, lipid and hydrogen, to optimize use in future space based and ground applications. o Education of the next generation of space flight investigators. o This proposal directly addresses the Space Biology Plan Element GL3: What are the key cellular and molecular systems necessary for biological organisms to thrive in space environment that can be discovered from 21st century bioinformatics tools (also Decadal survey page 71). o This application directly addresses three elements of the Human Research Roadmap on Risk of Radiation Carcinogenesis: Cancer 07: How can systems biology approaches be used to integrate research on the molecular, cellular, and tissue mechanisms of radiation damage to improve the prediction of the risk of cancer and to evaluate the effectiveness of countermeasures? How can epidemiology data and scaling factors support this approach? Cancer 08: What are the most effective medical or dietary countermeasures to mitigate cancer risks from exposure to SPE (solar particle events) and GCR (galactic cosmic rays)? What side effects should be tolerated versus mission risks? Can medical countermeasures that provide mitigation across multiple cancer types or across multiple radiation risk areas be identified? Cancer 09: Are there significant effects from other spaceflight factors that modify the carcinogenic risk from space radiation?

Task Progress:	Major Milestones: o The flight hardware has been built and validated o The Science Verification Test has been completed, and data analysis with report drafting is ongoing. Science Verification Test parameters requested by program. “For the Science Verification Test parameters, November 26 – January 31 (66 days) was allotted to perform the Science Verification Test. The exact duration of the Science Verification Test open to each of the was Principal Investigators (PIs) based on the requirements of their investigation (and current estimate of the Artemis I mission timeline). Same as shuttle and ISS, the objective of the Science Verification Test was to verify that your science and hardware together will enable you to be successful based on the science criteria set forth in the Experiment Requirements Document (ERD). Thus, to a large extent the work performed with Moonshot and the maturity of the hardware and science will better define what is needed to be done during the Science Verification Test time period.” SCIENCE VERIFICATION TEST Protocols. The Science Verification Test was performed in the Hammond lab, room 115, of building 15 the Research Building, at the Durham VA Medical Center. A loop inoculum of Chlamydomonas reinhardtii was placed into liquid Tris-acetate-phosphate media (1M Tris base 20 ml, phosphate K2HPO4/KH2PO4 buffer 1 ml, Hutner’s Trace metals 1 ml, 10 ml of solution A [NH4Cl, MgSP4.7H2O, CaCl2.2H2O], glacial acetic acid 1 ml pH 7 made up to 1 liter with distilled water, and exposed to white light. After a lag of ~21 days the Chlamydomonas grew robustly. TAP media plus 12 grams agar per liter was autoclaved and 10 ml poured into 100 mm diameter Petri dishes and single cell Omniplates. The dark green robustly growing Chlamydomonas culture was spotted onto the plates on Wednesday December 4. A pattern of 5 µl spots was arranged on each plate. The plates were parafilmed and allowed to dry at room temperature in a box containing red and blue light coming on contemporaneously for 6 hours out of a 24-hour cycle. The plates were plated in the flight hardware prototype and the Science Verification Test initiated Friday, December 6 at 12 noon midday. The Science Verification Test was completed on February 4, 2020 after 60 days. Omniplates were prepared as closely as possible as the round plates in the prototype, using the same batches of reagents and algae to make 16 spots. These Omniplates were exposed to the same light and temperature conditions, thereby providing a resource that allowed repetitive optical density measurements to define chlorophyll levels over time as an index of viability during the duration of the science verification test. Hardware. Bionetics custom hardware is an ultra-low power grow system to be used on EM-1. The system provides blue (~450 nm) and red (~660 nm) light to the experimental samples for continuously eight hours in every 24 hours. The hardware monitor and records the temperature of the experiment and the acceleration levels. Bionetics custom hardware consist of three module types: • Battery modules (two); • LED grow modules (three); • Control board (one). In summary, the flight hardware has been constructed and validated. The Science Verification Test has been completed with data analysis and report preparation ongoing. The hardware batteries had moderate remaining power at the end of the 66 days.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Hammond T, Birdsall H, Allen P, Breauz S, Giaever G, Nislow C. "Use of EM-1 biological return samples of Chlamydomonas reinhardtii to optimize biofuel production and astronaut radiation risk." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. ASGSR 2019 meeting program. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019